

Data Lakes: Concept, Architecture and Benefits

Data lakes capture, refine, and explore data in its raw form. Find out how they are shaping the future of data management and analytics.



With the advent of social media, IoT and other advancements in technology, a huge amount of data is getting generated. The concept of data lakes emerged in order to extract the maximum benefits from this data -- for enhanced adaptability and strong data analytics. A data lake is a storage space for storing heterogeneous data, both organised as well as unstructured. It improves the capture, refinement, and exploration of raw data within an enterprise. The data is kept in its original form, and the structure of the data is defined at the time of use, eliminating complex and costly data modelling.

Traditional decision support systems (DSS) are incapable of handling the huge amount of structured, unstructured, or semi-structured data generated by different resources. The data warehouse (DW) is the solution used by DSS. Here, the data is extracted, transformed and loaded (ETL processes) according to predefined schemas. However, the cost of a DW increases significantly

as the data size and complexity increase, and some information is lost through ETL processes.

Data lake architecture

Understanding data lake architecture can lead to more efficient data storage, faster processing, and better decision-making. The data lake architecture has two versions.

1. **Mono-zone:** This is a flat architecture that stores data in its native format. This architecture does not process data or record any user operations. It contains five data ponds that store data according to their characteristics:
 - Raw data ponds
 - Analog data ponds, to store analog types of data
 - Application data ponds, to store application data
 - Textual data ponds, to store text data
 - Archival data ponds, to store data that is no longer in use
2. **Multi-zone:** The multi-zone architecture has the following zones.

- Ingestion: Contains the raw data.
 - Storage: The ingested raw data is stored here.
 - Processing: Whenever data is processed, it is stored in the processing zone.
- This can be further divided into the following subzones.
- Batch processing
 - Real-time processing
- This zone controls data security, data quality, metadata management, and data life cycle.

Figure 1 defines the functional architecture of a data lake.

Data lake storage systems

Data lakes are changing the game for how businesses store and manage their data. Instead of siloed databases and spreadsheets, data lake systems allow you to store and access massive amounts of data in one place, giving you the flexibility to analyse it in real-time. They use different types of storage systems to achieve this. These are as follows.